

Subject Title (Theory): Cryptography and Information Security

Course code	Type of Course	Theory / Practical	Credits	Instruction hour per week	Duration of Exam	Formative Assessment Marks	Summative Assessment Marks	Total Marks
E5BCA1T2	Paper-2	Theory	04	04	3hrs	20	80	100

Course Outcomes (COs): At the end of the course, students will be able to:

- CO1. Identify basic security attacks and services
- CO2. Use symmetric and asymmetric key algorithms for cryptography
- CO3. Design a security solution for a given application
- CO4. Analyze Key Management techniques and importance of number Theory.
- CO5. Understanding of Authentication functions the manner in which Message Authentication Codes and Hash Functions works.
- CO6. To examine the issues and structure of Authentication Service and Electronic Mail Security

Total Hrs.: 60

Unit I: **15 hrs.**

Introduction: Security trends, The OSI Security Architecture, Security Attacks, Security Services and Security Mechanisms, A model for Network security.

Classical Encryption Techniques: Symmetric Cipher Modes, Substitute Techniques, Transposition Techniques, Rotor Machines, Stenography.

Unit II: **15 hrs.**

Block Cipher and Data Encryption Standards: Block Cipher Principles, Data Encryption Standards, the Strength of DES, Differential and Linear Crypt Analysis, Block Cipher Design Principles.

Advanced Encryption Standards: Evaluation Criteria for AES, the AES Cipher.

More on Symmetric Ciphers: Multiple Encryption, Triple DES, Block Cipher Modes of Operation, Stream Cipher and RC4.

Unit III: **15 hrs.**

Introduction to Number Theory: Prime Numbers, Fermat's and Euler's Theorem, Testing for Primality, The Chinese Remainder Theorem, Discrete logarithms.

Public Key Cryptography and RSA: Principles Public key crypto Systems, Diffie Hellman Key Exchange, the RSA algorithm, Key Management, , Elliptic Curve Arithmetic, Elliptic Curve Cryptography.

Unit IV: **15 hrs.**

Message Authentication and Hash Functions: Authentication Requirement, Authentication Function, Message Authentication Code, Hash Function, Security of Hash Function and MACs.

Hash and MAC Algorithm: Secure Hash Algorithm, Whirlpool, HMAC, CMAC.

Digital Signature: Digital Signature, Authentication Protocol, Digital Signature Standard.

Text Books:

1. William Stallings (2006), Cryptography and Network Security: Principles and Practice, 4th edition, Pearson Education, India.
2. William Stallings (2000), Network Security Essentials (Applications and Standards), Pearson Education, India.

References:

1. Charlie Kaufman (2002), Network Security: Private Communication in a Public World, 2nd edition, Prentice Hall of India, New Delhi.
2. Atul Kahate (2008), Cryptography and Network Security, 2nd edition, Tata Mc Grawhill, India.
3. Robert Bragg, Mark Rhodes (2004), Network Security: The complete reference, Tata Mc Grawhill, India.

Formative Assessment for Theory	
Assessment Type	Marks
Internal Assessment Test 1	05
Internal Assessment Test 2	05
Assignment	10
Total	20

Subject Title (Practical): Cryptography and Information Security Lab

Course code	Type of Course	Theory / Practical	Credits	Instruction hour per week	Duration of Exam	Formative Assessment Marks	Summative Assessment Marks	Total Marks
E5BCA1P2	Paper-2	Practical	02	04	3hrs	10	40	50

Course Outcomes (COs): At the end of the course (Practical), students will be able to:

CO1. Identify basic security attacks and services.

CO2. Use symmetric and asymmetric key algorithms for cryptography

CO3. Design a security solution for a given application.

Programs List:

1. Write a Java program to perform encryption and decryption using the following algorithms:
 - i. Ceaser Cipher
 - ii. Substitution Cipher
 - iii. Hill Cipher
2. Implement single and double transposition techniques
3. Perform encryption and decryption using following transposition techniques
 - i. Rail fence
 - ii. Row & Column Transformation
4. Write a Java program to implement the DES algorithm.
5. Write a Java program to implement the AES algorithm.
6. Write a Java program to implement the Tripple DES algorithm logic.
7. Write a JAVA program to implement the Blowfish algorithm logic
8. Write a JAVA program to implement the Rijndael algorithm logic
9. Write the RC4 logic in Java Using Java Cryptography, encrypt text “Hello world” using Blowfish Create your own key using Java key tool.
10. Write a Java program to implement RSA Algorithm.
11. Implement the Diffie-Hellman Key Exchange mechanism using HTML and JavaScript.
12. Calculate the message digest of a text using the SHA-1 algorithm in JAVA
13. Calculate the message digest of a text using the MD5 algorithm in JAVA
14. Implement the SIGNATURE SCHEME – Digital Signature Standard in Java